Neural networks for scientific discovery: hard-won insights.

Alexander G D G Matthews

Speaker bio



Undergraduate in physics, Cambridge

Clean tech startup, Oxford

PhD and postdoc in ML, Cambridge

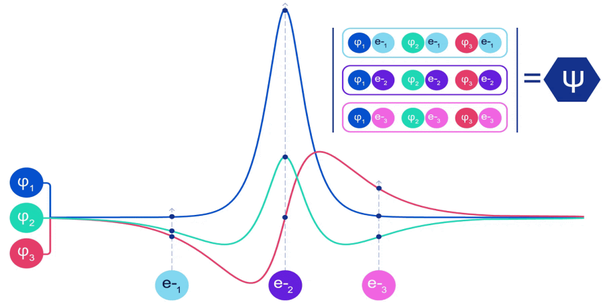
Eight years at Google DeepMind

Move to CuspAI

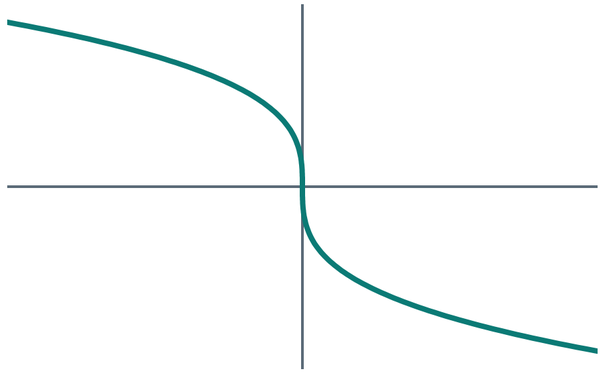
Please come and ask me about any of this!

Plan for the talk

Fermionic Neural Networks



Discovering equation blow-up solutions



Material discovery at CuspAI

cusp.ai

What is meant by "scientific discovery"?

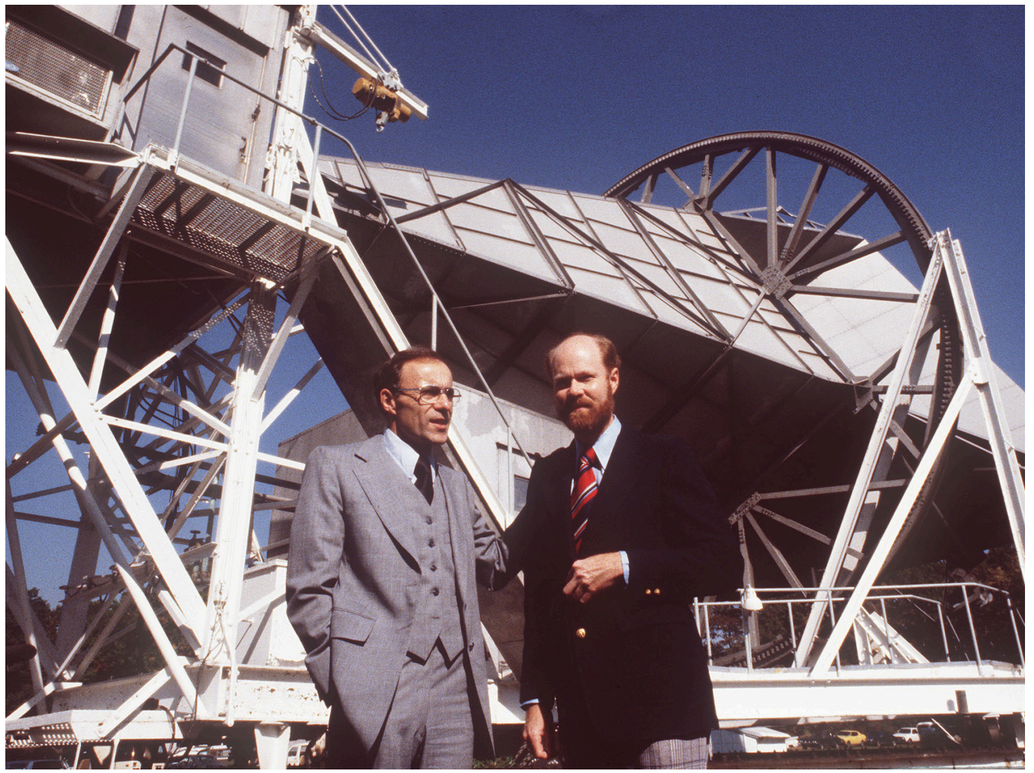
Definition needs to be broad to include everything in the talk!

Ideally would discover something about science that is previously unknown.

Allowed to include new mathematical results.

But the methodology is not considered discovery it is a means to an end.

Theme of the talk: even with good methods you need to be "on it" to discover things.



Penzias and Wilson at the Holmdel horn antenna

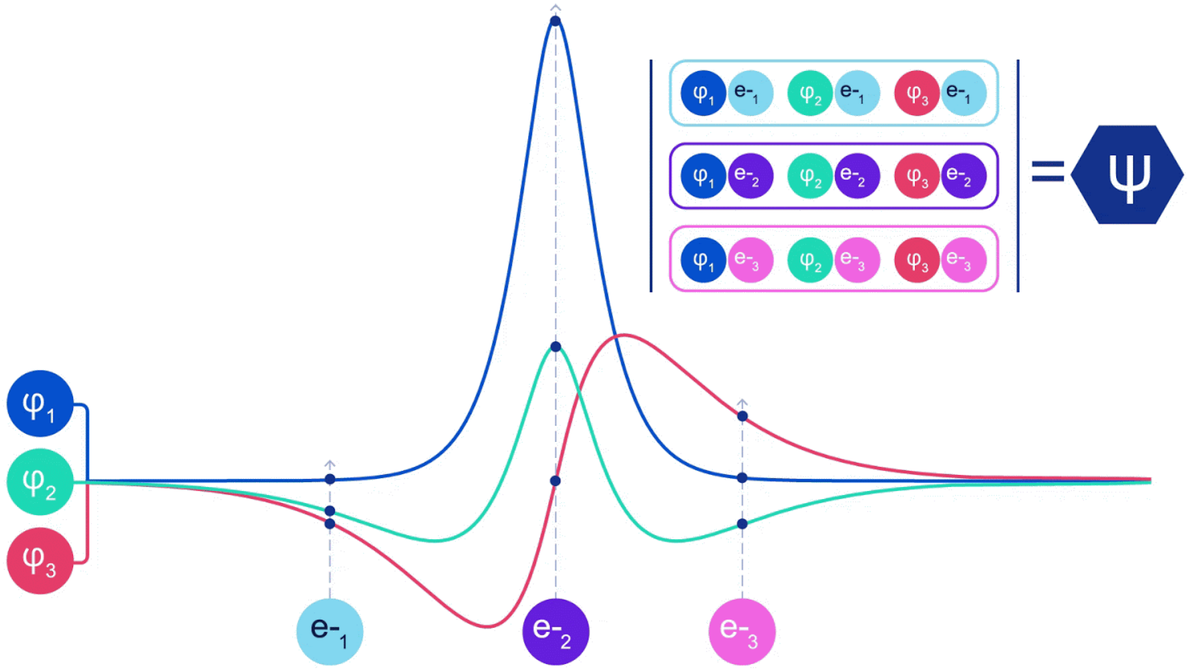
Three analogies for research strategy

Fermionic Neural Networks - *Starting a camp fire*

Discovering equation blow-up solutions - *Surfing a wave*

Material discovery at CuspAI - *White water rafting*

Fermionic Neural Networks



Ab initio solution of the many-electron Schrödinger equation with deep neural networks

Phys. Rev. Research 2020



David Pfau



James S Spencer



Matthew Foulkes

DeepMind

Imperial College London





Many electron Schrödinger equation

Schrödinger equation $\hat{H}\,\psi(\mathbf{x}_1, \ldots, \mathbf{x}_n) = E\,\psi(\mathbf{x}_1, \ldots, \mathbf{x}_n)$

Hamiltonian

$$\hat{H} = -\frac{1}{2}\sum_i \nabla_i^2 + \sum_{i>j}\frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} - \sum_{i,I}\frac{Z_I}{|\mathbf{r}_i - \mathbf{R}_I|} + \sum_{I>J}\frac{Z_I Z_J}{|\mathbf{R}_I - \mathbf{R}_J|}$$



“The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known, and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble.” P. A. M. Dirac, 1929

Electron anti-symmetry

Electron i

$$\mathbf{x}_i = \{ \underbrace{\mathbf{r}_i}_{\text{position}} , \underbrace{\sigma_i}_{\text{spin}} \}$$

Swap two electrons

$$\psi(\mathbf{x}_2, \mathbf{x}_1, \mathbf{x}_3, \ldots, \mathbf{x}_n) = -\psi(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \ldots, \mathbf{x}_n)$$

Anti-symmetry using determinants

$$\psi(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N) = \det \begin{pmatrix} f_1(\mathbf{x}_1) & f_2(\mathbf{x}_1) & \cdots & f_N(\mathbf{x}_1) \\ f_1(\mathbf{x}_2) & f_2(\mathbf{x}_2) & \cdots & f_N(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ f_1(\mathbf{x}_N) & f_2(\mathbf{x}_N) & \cdots & f_N(\mathbf{x}_N) \end{pmatrix}$$

Anti-symmetry using determinants

$$\psi(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N) = \det \begin{pmatrix} f_1(\mathbf{x}_1; \{\mathbf{x}_{/1}\}) & f_2(\mathbf{x}_1; \{\mathbf{x}_{/1}\}) & \cdots & f_N(\mathbf{x}_1; \{\mathbf{x}_{/1}\}) \\ f_1(\mathbf{x}_2; \{\mathbf{x}_{/2}\}) & f_2(\mathbf{x}_2; \{\mathbf{x}_{/2}\}) & \cdots & f_N(\mathbf{x}_2; \{\mathbf{x}_{/2}\}) \\ \vdots & \vdots & \ddots & \vdots \\ f_1(\mathbf{x}_N; \{\mathbf{x}_{/N}\}) & f_2(\mathbf{x}_N; \{\mathbf{x}_{/N}\}) & \cdots & f_N(\mathbf{x}_N; \{\mathbf{x}_{/N}\}) \end{pmatrix}$$

$f_j(\mathbf{x}_1; \{\mathbf{x}_{/1}\})$: an unconstrained function of $\mathbf{x}_1$, permutation invariant in all other electrons.

This is equivalent to permutation equivariance.

Idea dates back as far as Feynman and Cohen (1956) under the name "backflow".

Variational bound

Rayleigh quotient $$\lambda_{\min}(A) \leq \frac{x^T A\, x}{x^T x}$$

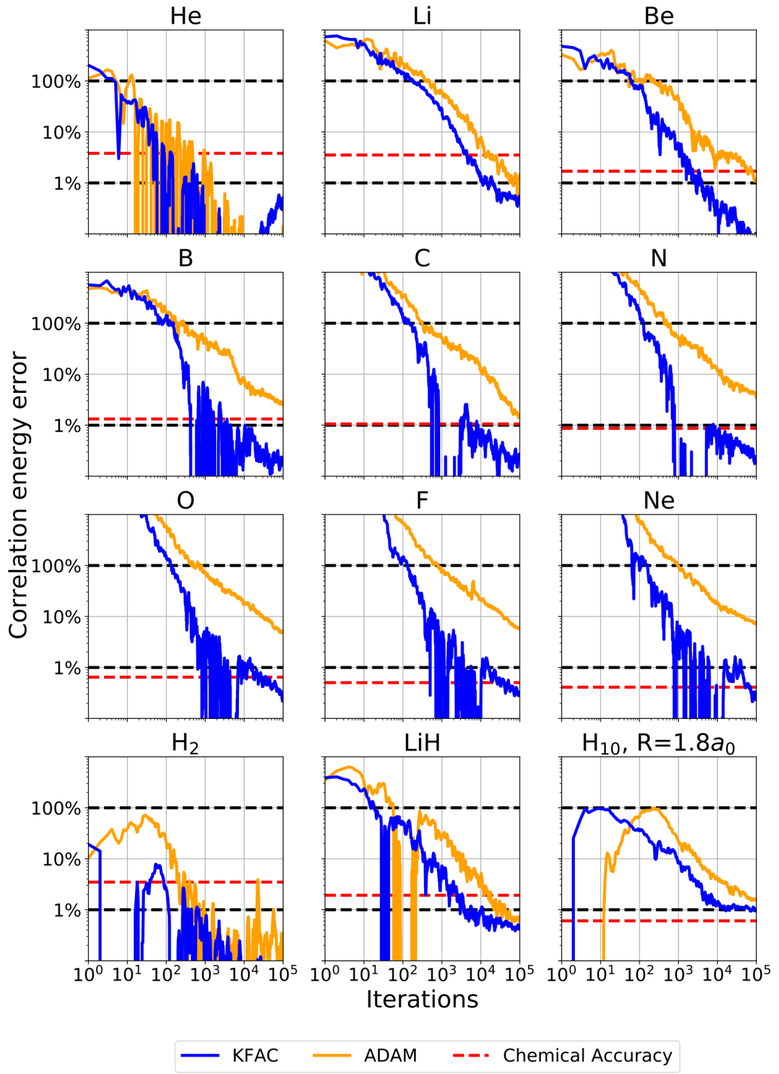
Variational bound $$E_0 \leq \frac{\langle \psi_\theta | \hat{H} | \psi_\theta \rangle}{\langle \psi_\theta | \psi_\theta \rangle}$$

High accuracy 2nd order optimization

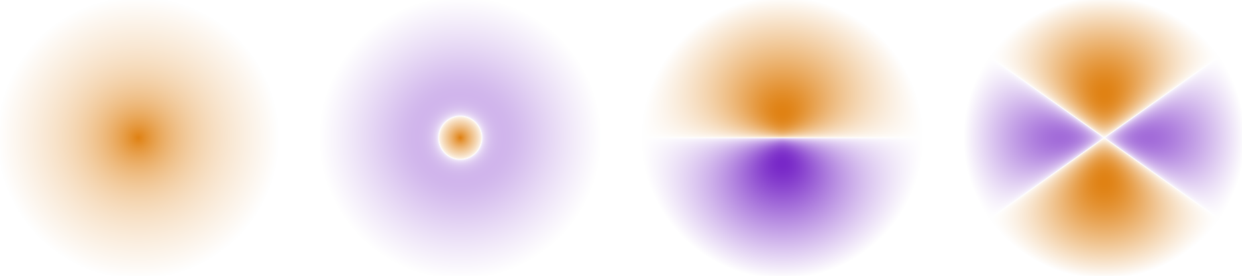
Second order optimizers significantly better than first order optimizers.

The additional accuracy was the difference between people caring or not.

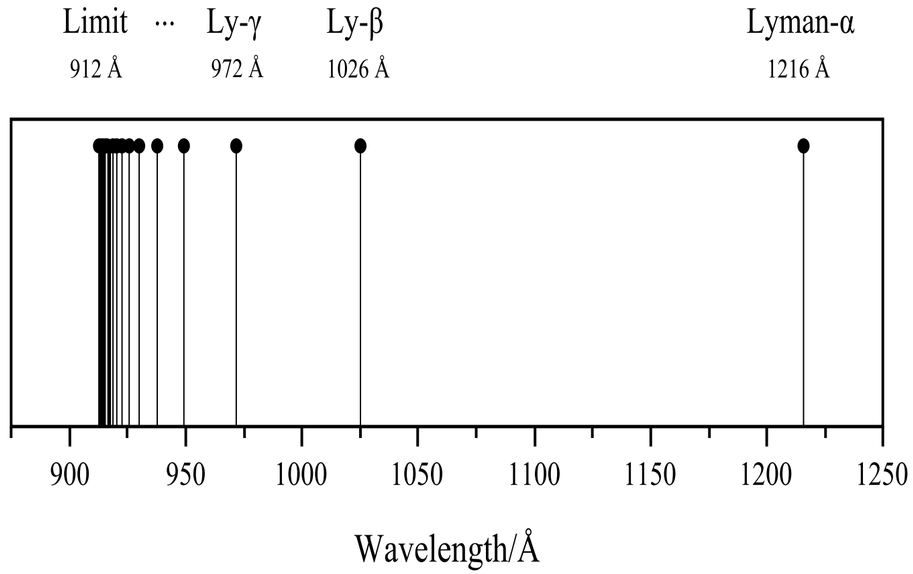
Need to use the KFAC optimizer of Martens and Grosse (2015) to scale number of parameters.



Excited states and spectra

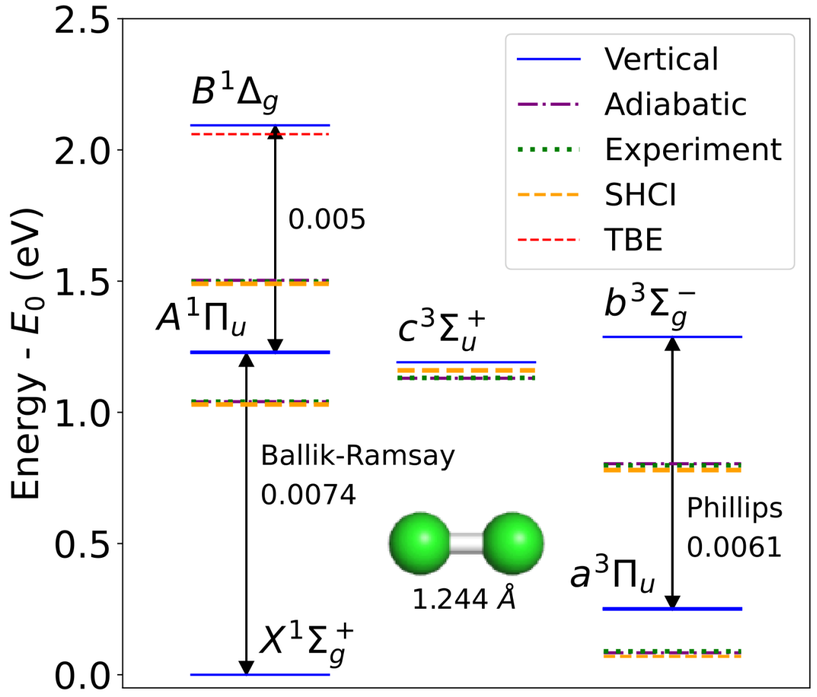


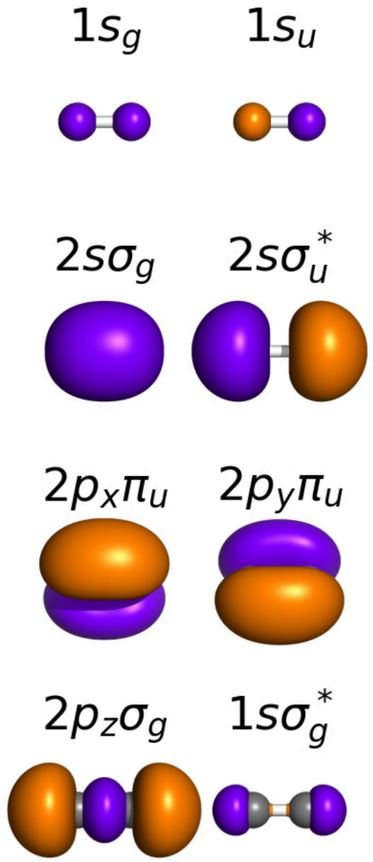
1s 2s 2p 3d



$$\hat{H}\psi = E\psi$$

Carbon dimer







Comet Lovejoy

Pfau et al., Science 385 (2024)

Discovering equation blow-up solutions

Discovery of Unstable Singularities

Yongji Wang[1, 2], Mehdi Bennani[3], James Martens[3], Sébastien Racanière[3], Sam Blackwell[3], Alex Matthews[3], Stanislav Nikolov[3], Gonzalo Cao-Labora[1, 4], Daniel S. Park[5], Martin Arjovsky[3, †], Daniel Worrall[3, †], Chongli Qin[3, †], Ferran Alet[3, †], Borislav Kozlovskii[3, †], Nenad Tomašev[3, †], Alex Davies[3], Pushmeet Kohli[3], Tristan Buckmaster[1,*], Bogdan Georgiev[3,*], Javier Gómez-Serrano[6,*], Ray Jiang[3,*] and Ching-Yao Lai[2,*]

[1]New York University, Department of Mathematics, New York, NY 10012, USA
[2]Stanford University, Department of Geophysics, Stanford, CA 94305, USA
[3]Google DeepMind, London, N1C 4DJ, UK
[4]École Polytechnique Fédérale de Lausanne, Institute of Mathematics, Lausanne, VA 1015, Switzerland
[5]Google DeepMind, New York, NY 10011, USA
[6]Brown University, Department of Mathematics, Providence, RI 02912, USA

2025 https://arxiv.org/pdf/2509.14185



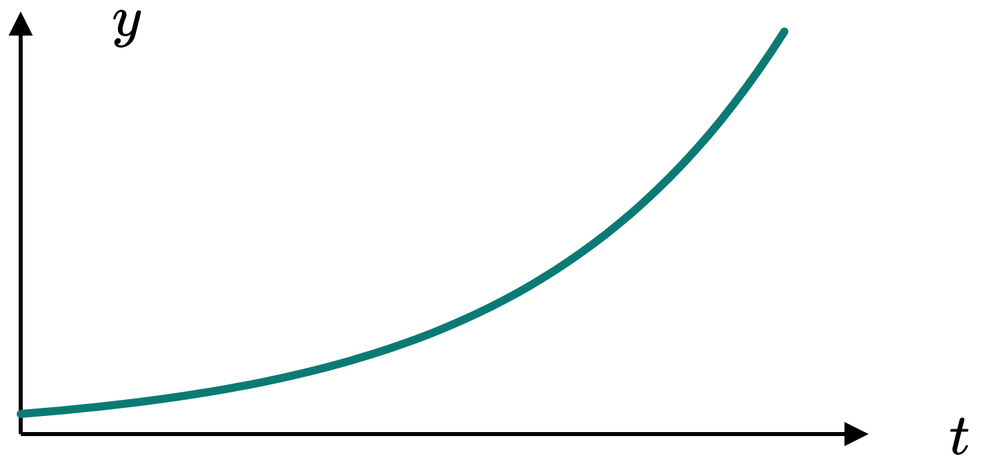


Nonlinear blow-up

Exponential growth

$$\frac{dy}{dt} = ky$$

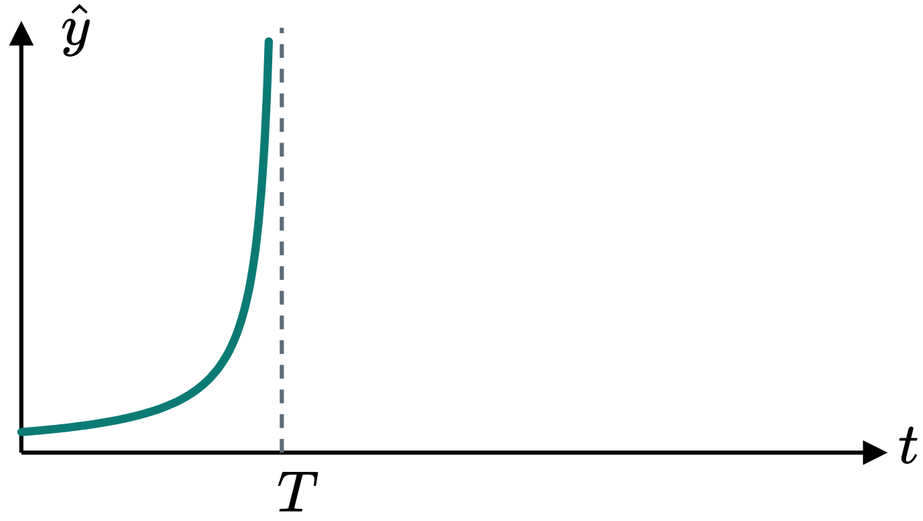
$$y = Ae^{kt}$$



Finite time blow-up

$$\frac{d\hat{y}}{dt} = \kappa \hat{y}^2$$

$$\hat{y} = \frac{1}{\kappa(T - t)}$$

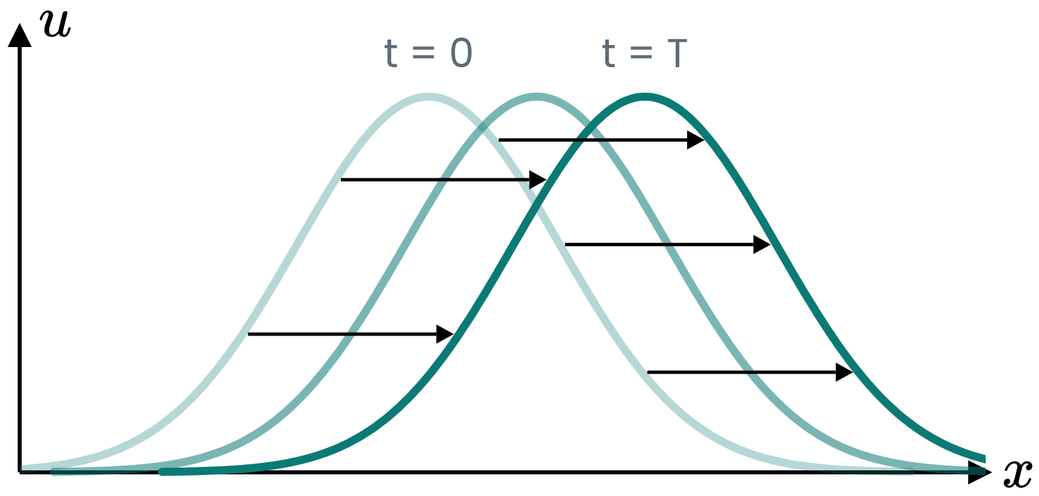


Burgers equation

One-way wave: constant speed c

$$\frac{\partial u}{\partial t} + c\,\frac{\partial u}{\partial x} = 0$$

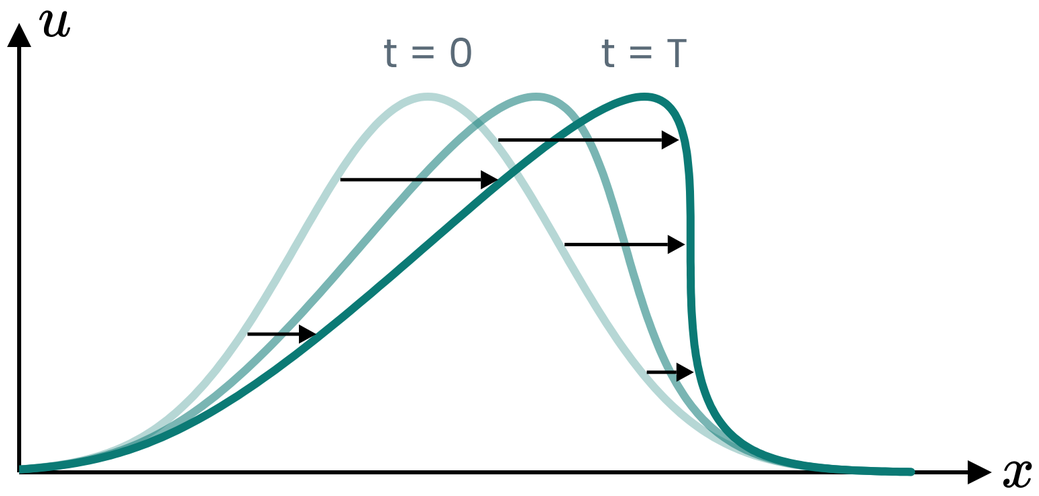
$$u = f(x - ct)$$



Burgers: the speed is the height u

$$\frac{\partial u}{\partial t} + u\,\frac{\partial u}{\partial x} = 0$$

$$u = f(x - ut)$$



From the wave to the shock

Follow one point of the wave. Its steepness $s = -\partial u/\partial x$ obeys

$$\frac{ds}{dt} = s^2$$

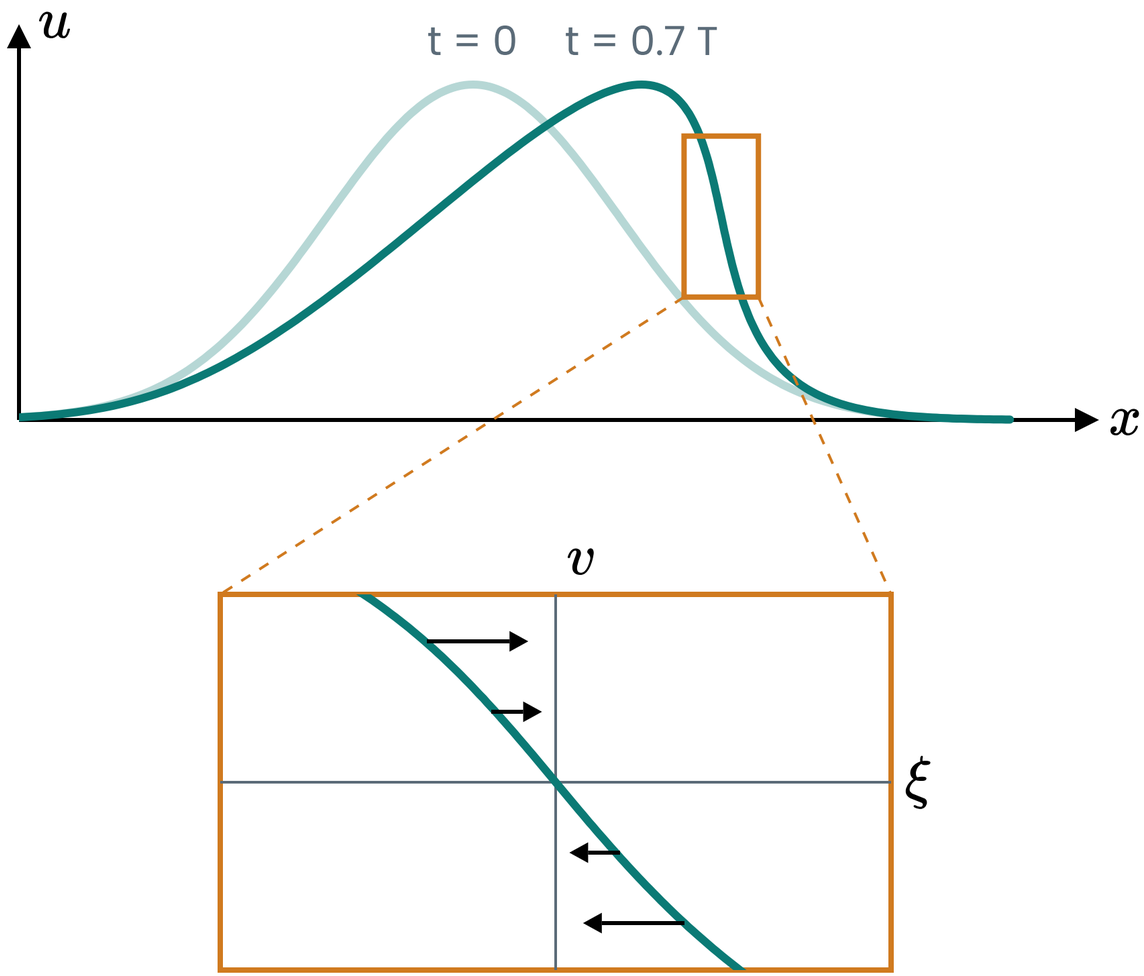
the blow-up from before: the steepest point turns vertical first, at time T.

Ride along with that point at its speed c:

$v = u - c, \xi = x - a_* - ct$

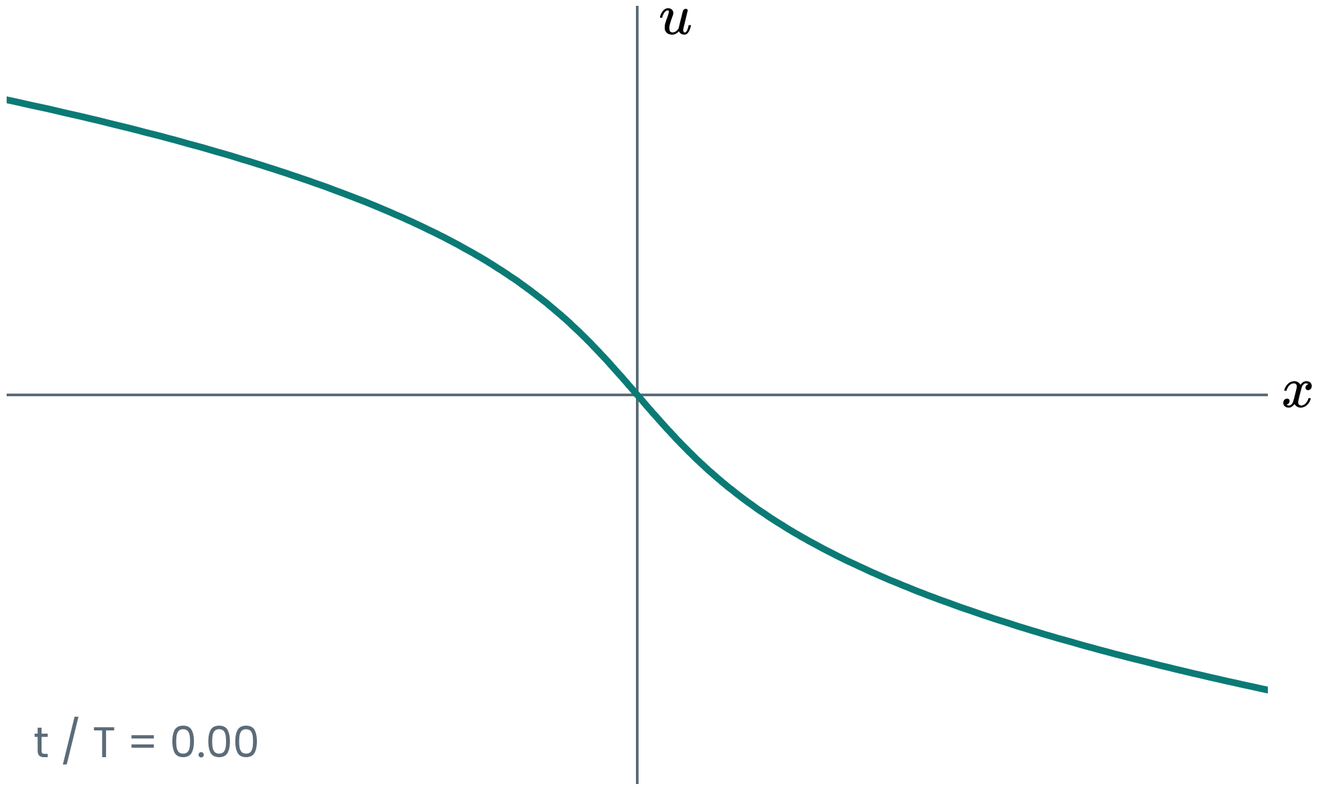
$$\frac{\partial v}{\partial t} + v\,\frac{\partial v}{\partial \xi} = 0$$

Same equation. Now the fluid squeezes in from both sides.



Self-similar Burgers shock formation

$$\frac{\partial u}{\partial t} + u\,\frac{\partial u}{\partial x} = 0, \qquad u(x,t) = (T-t)^{1/2}\,\Psi_1\left(\frac{x}{(T-t)^{3/2}}\right), \qquad X = -\Psi_1(X) - \Psi_1(X)^3$$



Self-similar Burgers PDE

Burgers equation

$$\frac{\partial u}{\partial t} + u\,\frac{\partial u}{\partial x} = 0$$

Self-similar ansatz

$$u(x,t) = (T-t)^{\lambda}\,U(y), \qquad y = \frac{x}{(T-t)^{1+\lambda}}$$

Substituting...

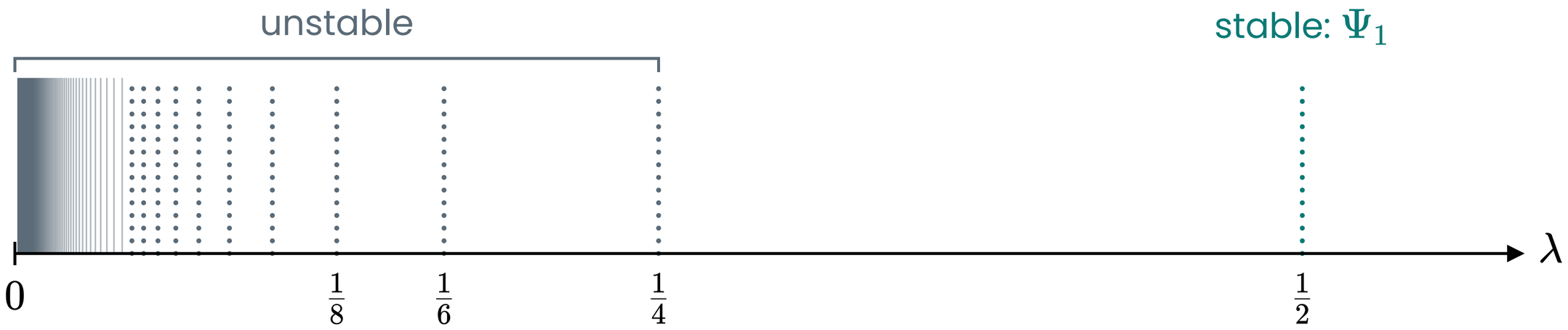
$$-\lambda\,U + \Big((1+\lambda)\,y + U\Big)\,\frac{\partial U}{\partial y} = 0$$

Solving self-similar Burgers PDE

Linear ODE for $y(U)$ $$\lambda U \frac{dy}{dU} = (1+\lambda)\, y + U$$

Implicit solution $$y = -U - U|U|^{1/\lambda}, \qquad U(-2) = 1$$

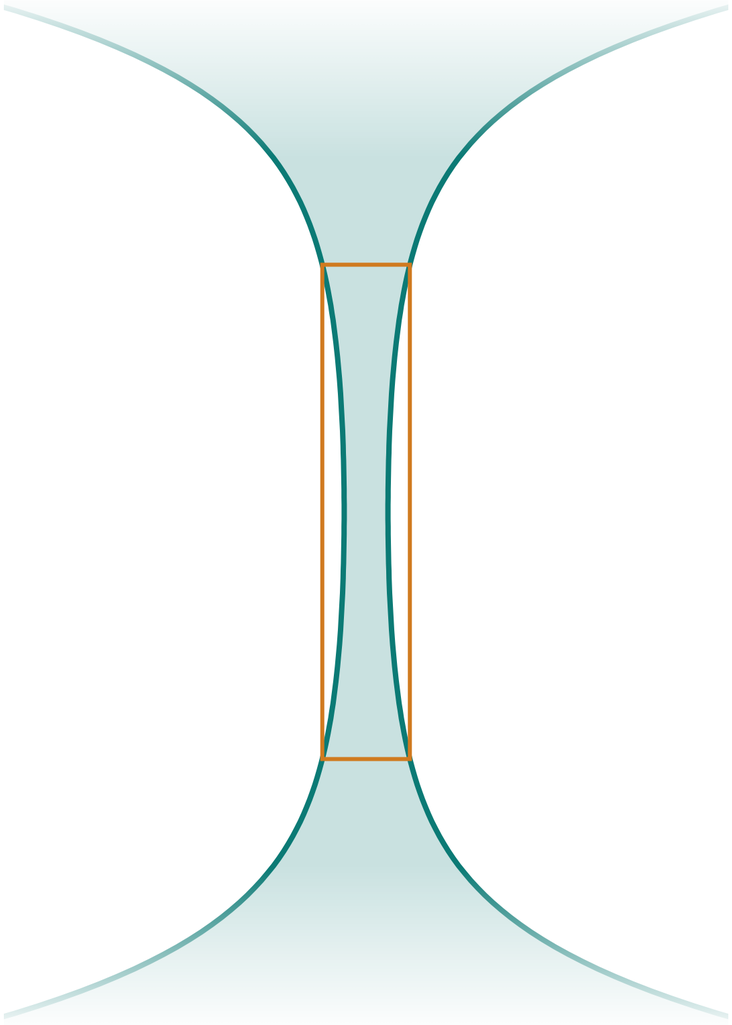
Smooth iff. odd power $$1 + \frac{1}{\lambda} = 3, 5, 7, \ldots \quad \Longrightarrow \quad \lambda = \frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \ldots$$



Same pattern: viscous pinch-off

$$h(x,t) = (T-t)\,\phi\!\left(\frac{x}{(T-t)^{\beta}}\right)$$

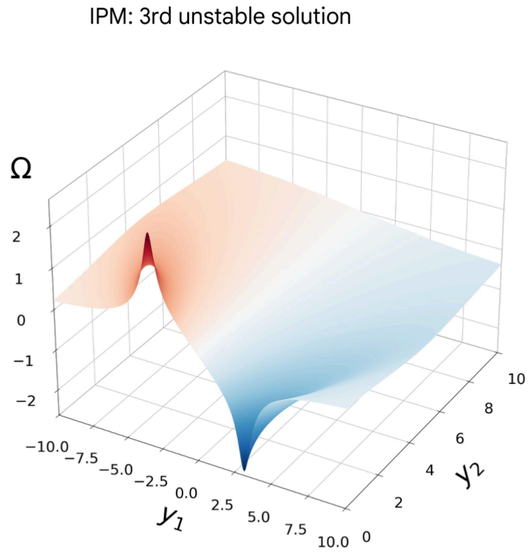
Again a discrete family, $\beta = 0.175,\ 0.045,\ 0.019,\ \ldots$,
and only the first is stable.

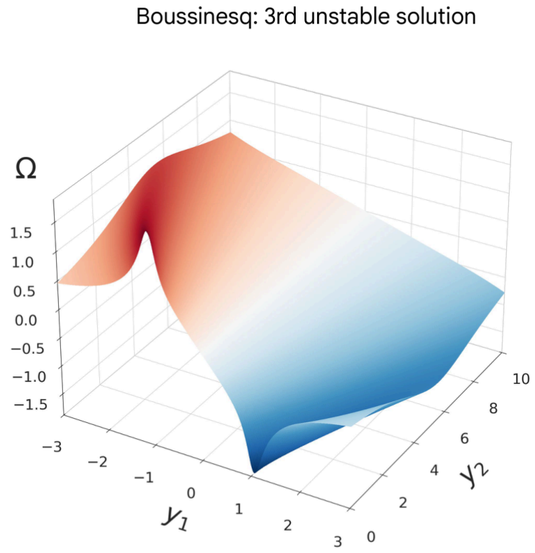


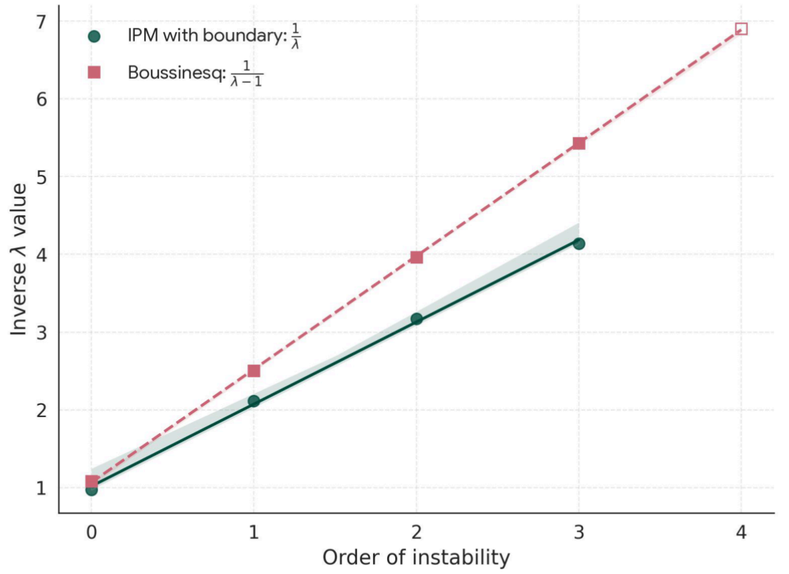
Fluid-like nonlinear PDEs

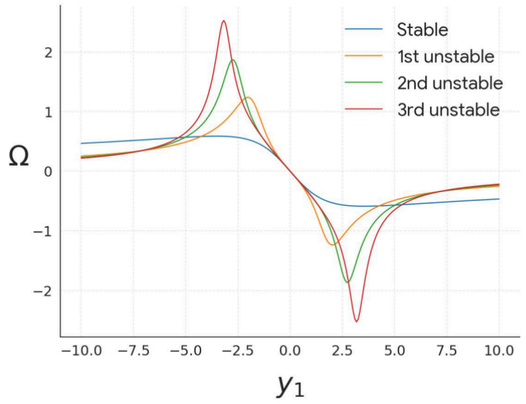
Boussinesq 2D	$\frac{\partial u}{\partial t} + (u \cdot \nabla) u + \nabla p = (0, \theta)$ $\frac{\partial \theta}{\partial t} + u \cdot \nabla \theta = 0, \qquad \nabla \cdot u = 0$	Buoyant flow: temperature θ is carried along and pushes fluid up. With a wall it mirrors axisymmetric 3D Euler.
CCF 1D	$\frac{\partial \theta}{\partial t} + (H\theta) \frac{\partial \theta}{\partial x} = 0$	θ carried by a nonlocal velocity, its Hilbert transform $H\theta$. A 1D model of 3D Euler and Navier–Stokes.
IPM with boundary 2D	$\frac{\partial \rho}{\partial t} + u \cdot \nabla \rho = 0$ $u = -\nabla p - (0, \rho), \qquad \nabla \cdot u = 0$	Density ρ carried by flow in a porous medium. Darcy's law under gravity replaces Newton's second law.

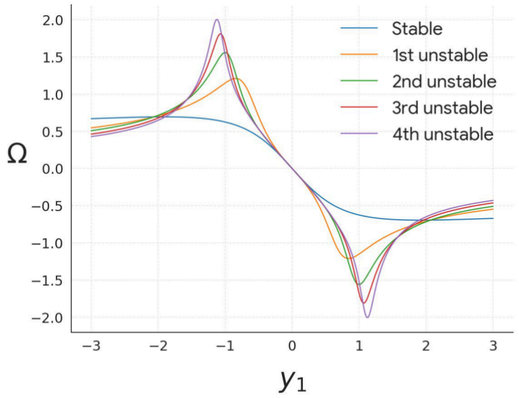
Discovered unstable singularities





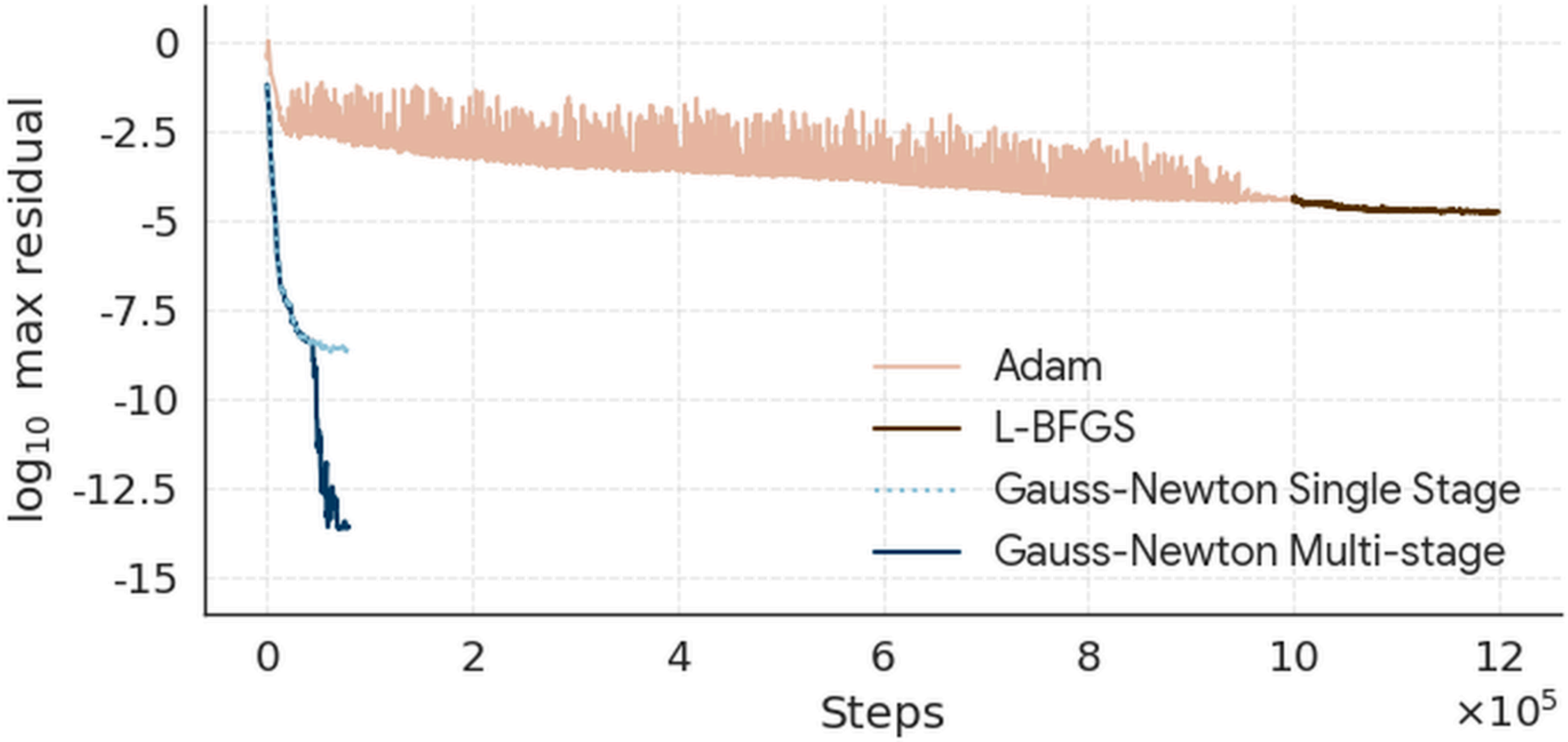






Lambda	CCF	IPM with boundary	Boussinesq
Stable	1.1807776628998	1.028572276 0222	1.920559 9746927
1st unstable	0.6057337012032	0.472129736 2414	1.399096 1221852
2nd unstable	0.471324 8638620	0.314962 0267088	1.252348 1636489
3rd unstable		0.241560 4743989	1.184250 0861997
4th unstable			1.1448833556188

High accuracy 2nd order optimization



Self-similar incompressible Euler blow-up?

Newton's second law

$$\underbrace{\frac{\partial u}{\partial t} + (u \cdot \nabla)\, u}_{\text{(Mass times) acceleration}} = \underbrace{-\nabla p}_{\text{Pressure force}}$$

Incompressibility

$$\nabla \cdot u = 0$$

Elgindi (2019): exactly self-similar blow-up, but only $C^{1,\alpha}$ (not smooth) and infinite energy.

Elgindi, Ghoul & Masmoudi (2019): *locally* self-similar, so finite energy; still only $C^{1,\alpha}$.

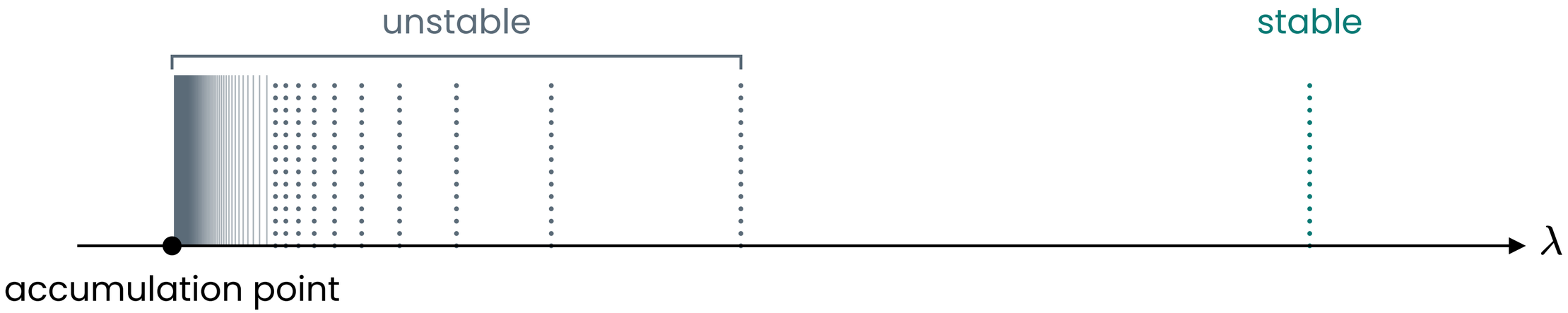
Luo & Hou (2014); Chen & Hou (2022): smooth, nearly self-similar, but needs a boundary.

OpenAI (2026) recently claimed a blow-up without boundary; no claim of self-similarity.

Self-similar incompressible Euler blow-up?

Question: Can we find a *smooth, locally-self-similar, finite-energy without boundary* solution?

Still open, despite many attempts, both numerical and algebraic.



If the accumulation point $\lambda^* < -1/2$ this would likely be transferrable to a Navier-Stokes blow-up which includes viscosity.

Constantin, Ignatova & Vicol, arXiv:2602.17570 (2026)

Navier-Stokes blow-up

Newton's second law

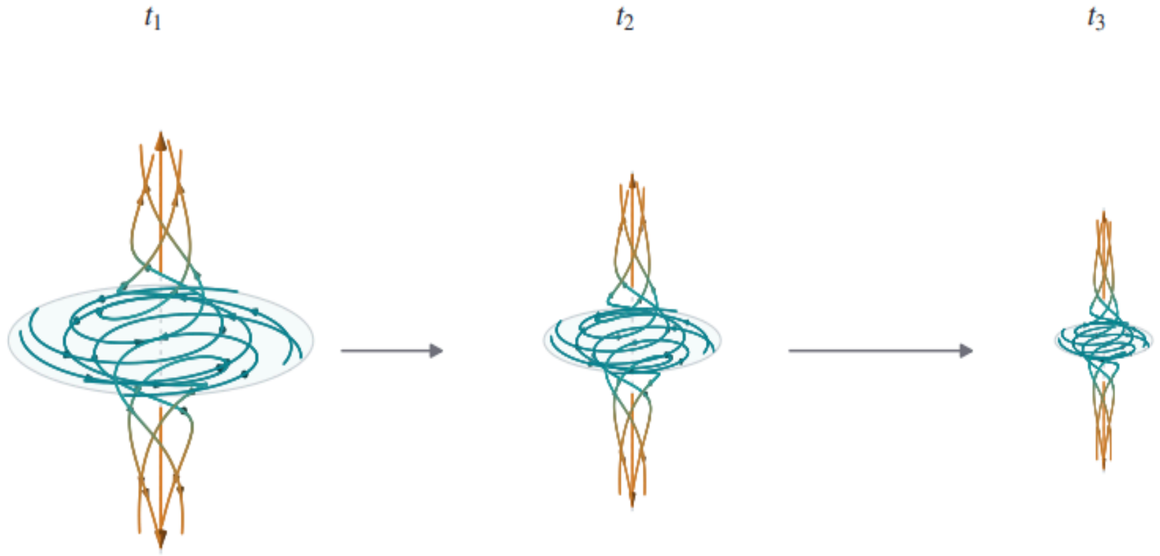
$$\underbrace{\frac{\partial u}{\partial t} + (u \cdot \nabla)\, u}_{\text{(Mass times) acceleration}} = \underbrace{-\nabla p}_{\text{Pressure force}} + \underbrace{\nu \nabla^2 u}_{\text{Viscous force}} + \underbrace{f}_{\text{External force}}$$

Incompressibility

$$\nabla \cdot u = 0$$

Related but distinct from self-similar construction

Axisymmetric leading core



Cylindrical polar coordinates

$\mathbf{x} = (r\cos\theta,\ r\sin\theta,\ z)$

$\tau = T - t$: time to blow-up

$$u_r = \tau^{-1/2}\, F_r\big(r/\tau^{1/2},\ z/\tau^{1/2-h}\big)$$
$$u_\theta = \tau^{-1/2-h}\, F_\theta\big(r/\tau^{1/2},\ z/\tau^{1/2-h}\big)$$
$$u_z = \tau^{-1/2-h}\, F_z\big(r/\tau^{1/2},\ z/\tau^{1/2-h}\big)$$
$$p = \tau^{-1-2h}\, P\big(r/\tau^{1/2},\ z/\tau^{1/2-h}\big)$$

Fixed $0 < h < \frac{1}{100}$: radius $\sim \tau^{1/2}$, length $\sim \tau^{1/2-h}$, so the core gets ever more slender.

OpenAI, Finite time blowup for Navier–Stokes (2026), §§3.1, 4.1

Related but distinct from self-similar construction

Self-similar, $\lambda = -\frac{1}{2}$

$$u = \tau^{-1/2}\, U\!\left(\frac{x}{\tau^{1/2}}\right), \qquad \tau = T - t$$

Leading core

$$\begin{aligned} u_r &= \tau^{-1/2}\, F_r\big(r/\tau^{1/2},\ z/\tau^{1/2-h}\big) \\ u_\theta &= \tau^{-1/2-h}\, F_\theta\big(r/\tau^{1/2},\ z/\tau^{1/2-h}\big) \\ u_z &= \tau^{-1/2-h}\, F_z\big(r/\tau^{1/2},\ z/\tau^{1/2-h}\big) \\ p &= \tau^{-1-2h}\, P\big(r/\tau^{1/2},\ z/\tau^{1/2-h}\big) \end{aligned}$$

Fixed $0 < h < \frac{1}{100}$: radius $\sim \tau^{1/2}$, length $\sim \tau^{1/2-h}$, so the core gets ever more slender.

Setting $h = 0$ gives exact self-similarity with $\lambda = -\frac{1}{2}$.

Leading core only: corrections, including non-axisymmetric oscillations, complete the flow.

OpenAI, Finite time blowup for Navier–Stokes (2026), §§3.1, 4.1

Material discovery at CuspAI

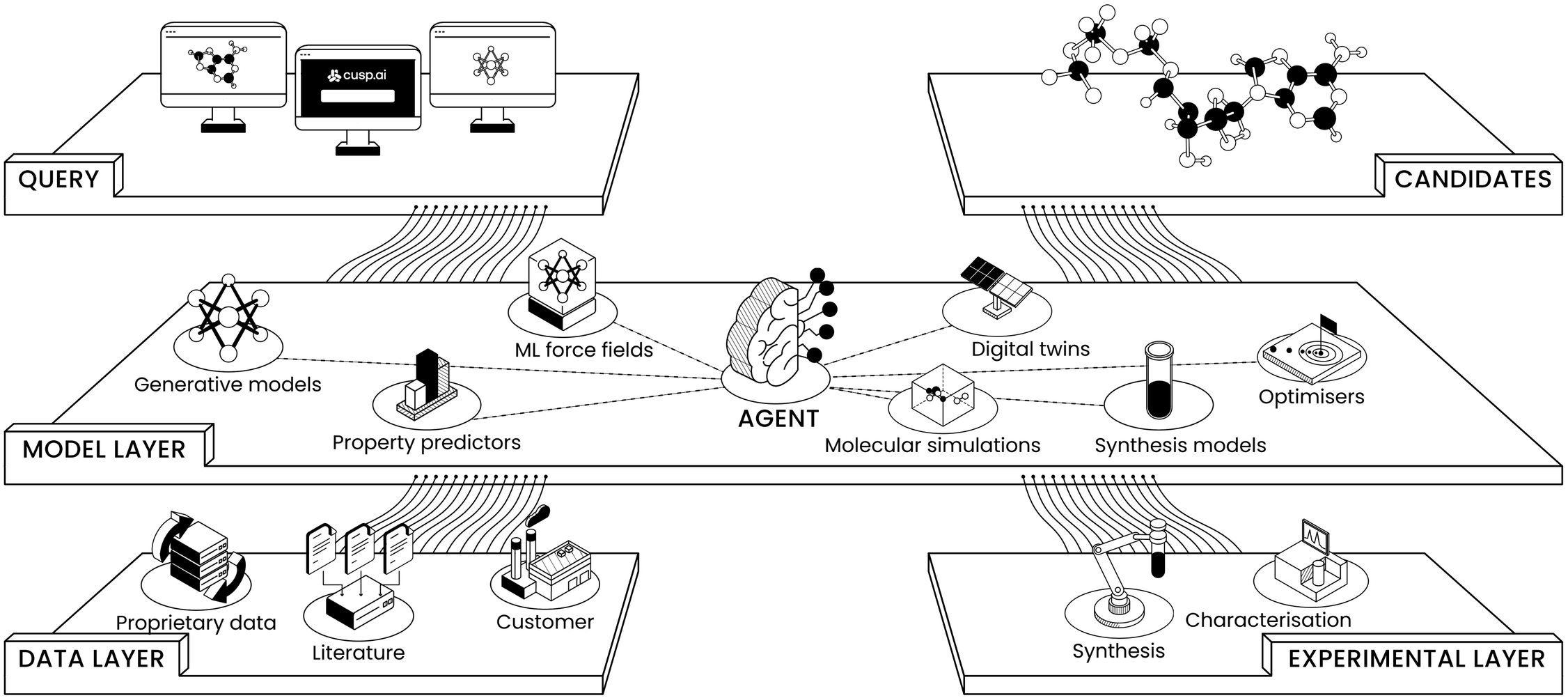




Our platform

Dim the rest

Clear



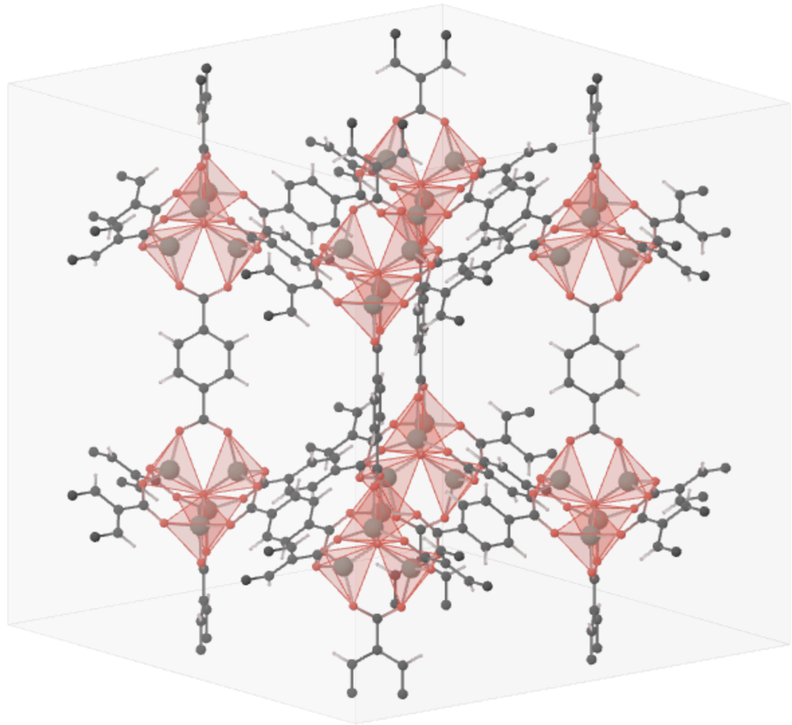
Metal Organic Frameworks

A MOF is a crystal of metal clusters joined by organic linkers. It is riddled with nanoscale pores, giving an enormous internal surface for molecules to stick to.

2025 Nobel Prize in Chemistry: Kitagawa, Robson and Yaghi.

Potential applications:

- Direct air capture (DAC) of CO_2
- Water purification, e.g. removing PFAS
- Harvesting drinking water from desert air



H_{96} C_{192} O_{104} Zn_{32}

Li, Eddaoudi, O'Keeffe & Yaghi, *Nature* 402, 276 (1999)

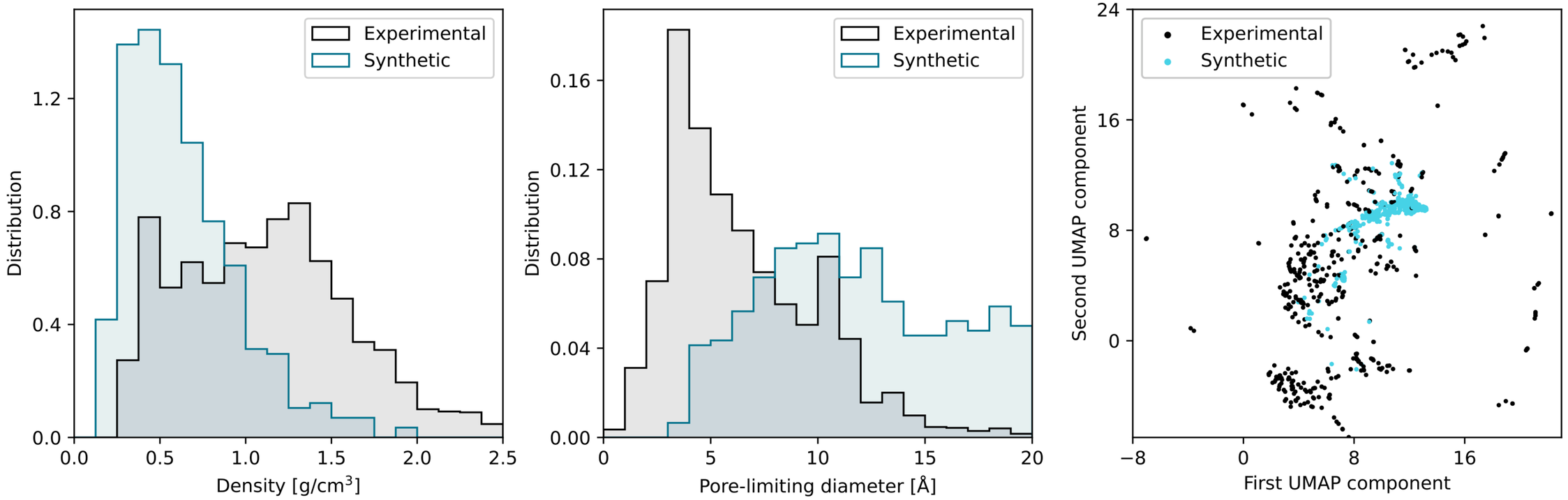
The Open DAC 2025 Dataset for Sorbent Discovery in Direct Air Capture

Anuroop Sriram[1,*], **Logan M. Brabson**[2,*], **Xiaohan Yu**[2,*], **Sihoon Choi**[2], **Kareem Abdelmaqsoud**[5], **Elias Moubarak**[4], **Pim de Haan**[4], **Sindy Löwe**[4], **Johann Brehmer**[4], **John R. Kitchin**[5], **Max Welling**[4], **C. Lawrence Zitnick**[1], **Zachary Ulissi**[1], **Andrew J. Medford**[2], **David S. Sholl**[3]

[1]FAIR at Meta, [2]School of Chemical and Biomolecular Engineering, Georgia Institute of Technology, [3]University of Tennessee-Oak Ridge Innovation Institute, Oak Ridge National Laboratory, [4]CuspAI, [5]Department of Chemical Engineering, Carnegie Mellon University

Nature Chemical Engineering 2026

Improved data diversity with generative modelling

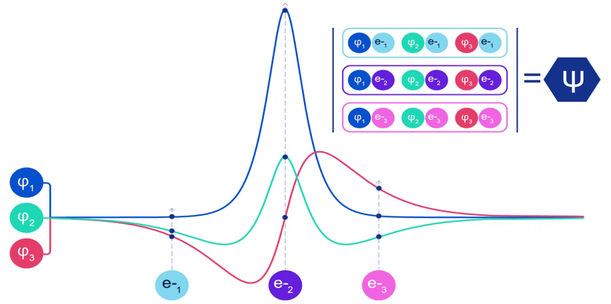


Sriram et al. (2025), Fig. S11

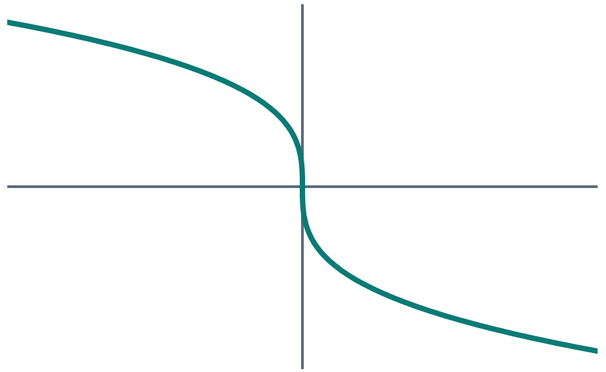
Conclusions and themes

Review of the talk

Fermionic Neural Networks



Discovering equation blow-up solutions



Material discovery at CuspAI

cusp.ai

Three analogies for research strategy

Fermionic Neural Networks - *Starting a camp fire*

Discovering equation blow-up solutions - *Surfing a wave*

Material discovery at CuspAI - *White water rafting*



Increasing need to work in larger teams.

Increasing need to plan for external forces.

Real world data is increasingly the bottleneck

Agentic AI is making coding and math increasingly fast.

Whilst access to HPC is expensive, I would argue it is not the bottleneck.

The physical world is extremely complex and models:

1) Are imperfect despite best practice.
2) Operate under a set of assumptions and at a certain scale.

The most impactful problems are now open to those who close these gaps.

alexggmatthews.github.io



References

Introduction and Fermionic Neural Networks

1. A. A. Penzias, R. W. Wilson, A measurement of excess antenna temperature at 4080 Mc/s, *Astrophys. J.* 142, 419 (1965).
2. D. Pfau, J. S. Spencer, A. G. D. G. Matthews, W. M. C. Foulkes, Ab initio solution of the many-electron Schrödinger equation with deep neural networks, *Phys. Rev. Research* 2, 033429 (2020).
3. R. P. Feynman, M. Cohen, Energy spectrum of the excitations in liquid helium, *Phys. Rev.* 102, 1189 (1956).
4. D. Luo, B. K. Clark, Backflow transformations via neural networks for quantum many-body wave functions, *Phys. Rev. Lett.* 122, 226401 (2019).
5. J. Hermann, Z. Schätzle, F. Noé, Deep-neural-network solution of the electronic Schrödinger equation, *Nat. Chem.* 12, 891 (2020).
6. P. A. M. Dirac, Quantum mechanics of many-electron systems, *Proc. R. Soc. Lond. A* 123, 714 (1929).
7. J. Martens, R. Grosse, Optimizing neural networks with Kronecker-factored approximate curvature, *ICML* (2015).
8. D. Pfau, S. Axelrod, H. Sutterud, I. von Glehn, J. S. Spencer, Accurate computation of quantum excited states with neural networks, *Science* 385, eadn0137 (2024).

Discovering equation blow-up solutions

9. C. Collot, T.-E. Ghoul, N. Masmoudi, Singularity formation for Burgers equation with transverse viscosity, arXiv:1803.07826 (2018).
10. Y. Wang, C.-Y. Lai, J. Gómez-Serrano, T. Buckmaster, Asymptotic self-similar blow-up profile for three-dimensional axisymmetric Euler equations using neural networks, *Phys. Rev. Lett.* 130, 244002 (2023).
11. J. Eggers, M. A. Fontelos, The role of self-similarity in singularities of PDEs, arXiv:0812.1339 (2008).

References (continued)

Discovering equation blow-up solutions (continued)

12. A. Córdoba, D. Córdoba, M. A. Fontelos, Formation of singularities for a transport equation with nonlocal velocity, *Ann. of Math.* 162, 1377 (2005).
13. Y. Wang et al., Discovery of unstable singularities, arXiv:2509.14185 (2025).
14. P. Constantin, M. Ignatova, V. Vicol, On putative self-similarity for incompressible 3D Euler, arXiv:2602.17570 (2026).
15. T. M. Elgindi, Finite-time singularity formation for $C^{1,\alpha}$ solutions to the incompressible Euler equations on $\mathbb{R}^3$, *Ann. of Math.* 194, 647 (2021).
16. T. M. Elgindi, T.-E. Ghoul, N. Masmoudi, On the stability of self-similar blow-up for $C^{1,\alpha}$ solutions to the incompressible Euler equations on $\mathbb{R}^3$, *Camb. J. Math.* 9, 1035 (2022).
17. G. Luo, T. Y. Hou, Potentially singular solutions of the 3D axisymmetric Euler equations, *PNAS* 111, 12968 (2014).
18. J. Chen, T. Y. Hou, Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data I: Analysis, arXiv:2210.07191 (2022).
19. OpenAI, Finite time blowup for Navier–Stokes (2026).

Material discovery at CuspAI

20. H. Li, M. Eddaoudi, M. O'Keeffe, O. M. Yaghi, Design and synthesis of an exceptionally stable and highly porous metal–organic framework, *Nature* 402, 276 (1999).
21. A. Sriram et al., The Open DAC 2025 dataset for sorbent discovery in direct air capture, *Nature Chem. Eng.* (2026); arXiv:2508.03162.